

Challenge Day Rules

Race the course. Protect the robot. Respect the room.

Engineers test carefully, observe carefully, and improve carefully.

PROTECT

PROTECT THE ROBOT

- Carry carefully
- No throwing or pushing
- Do not grab it mid-run

RESPECT

RESPECT TEAMS

- One driver at a time
- Do not touch other robots
- No blocking the course

SAFETY

SAFETY FIRST

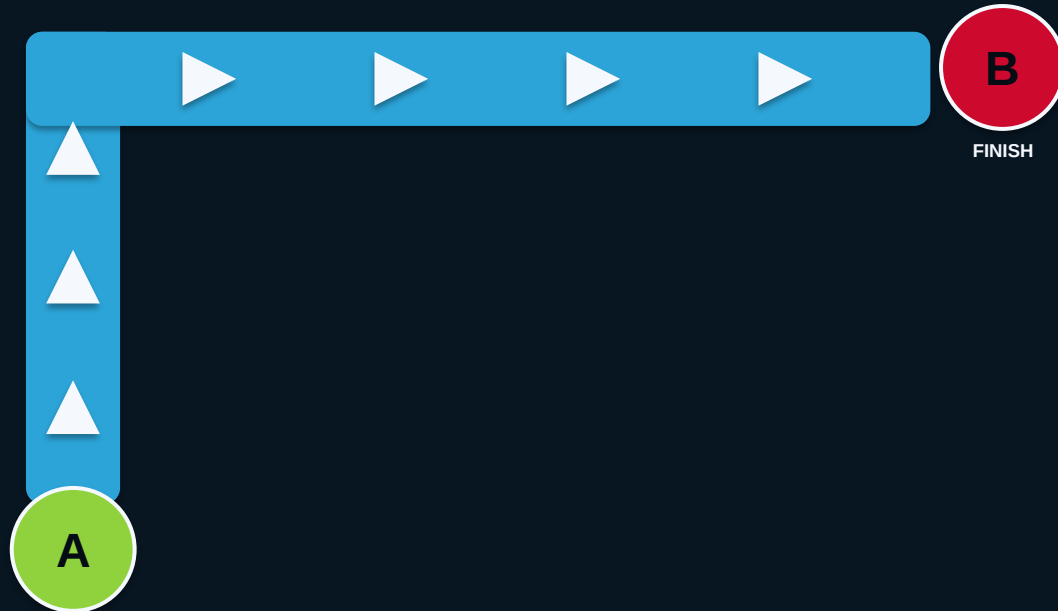
- STOP means freeze
- Listen to instructors
- No intentional crashes

Robot Timeout

Misuse = robot removed from the activity.

Track 1: First Turn

Point A to Point B with one 90° turn.



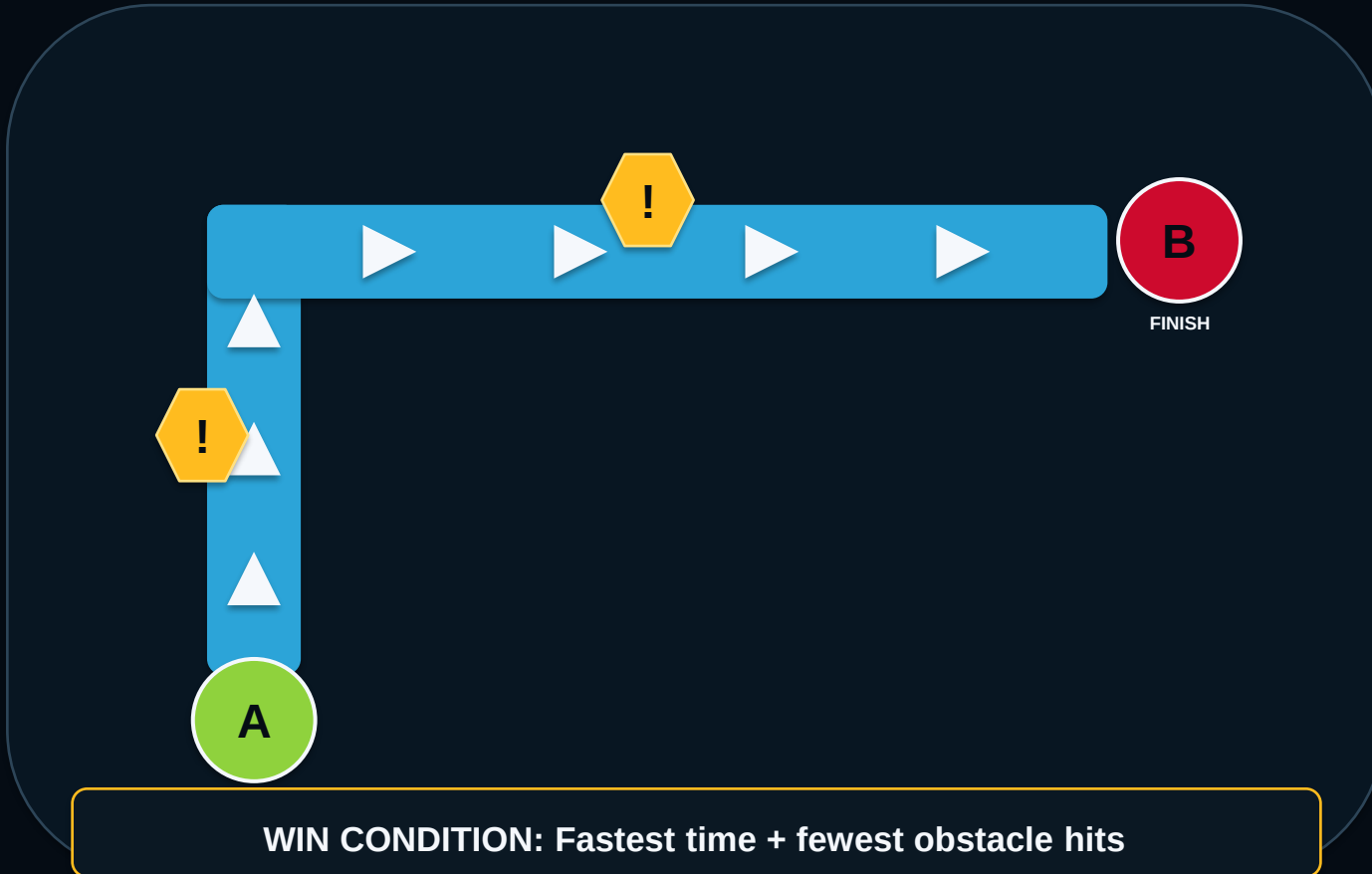
WIN CONDITION: Fastest time

GOAL + SKILLS

- 1 Mission**
Get from START A to FINISH B.
- 2 Course**
One turn. No obstacles. No walls.
- 3 Test**
Forward movement, timed turning, and planning.
- 4 Improve**
Change one command at a time, then run again.

Track 2: Dodge the Obstacles

Same idea, but now the path has objects in the way.

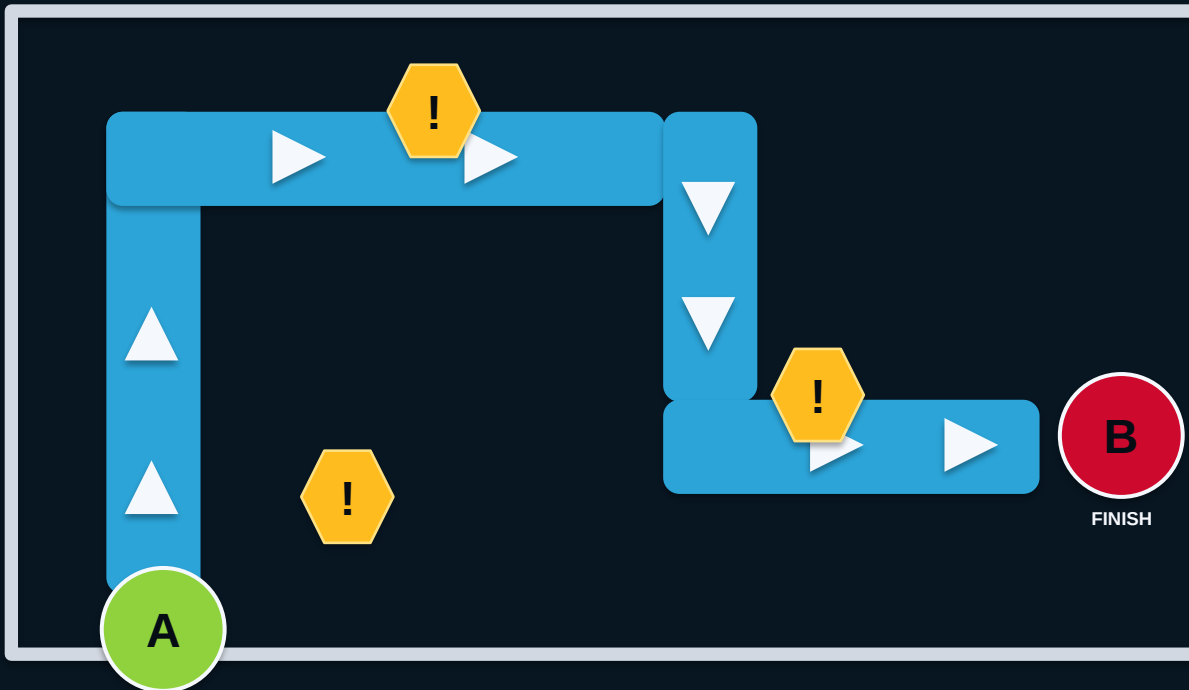


GOAL + RULES

- 1 Mission**
Reach Point B while avoiding obstacles.
- 2 Course**
Obstacles added. No walls yet.
- 3 Penalty**
Obstacle hits count against your run.
- 4 Strategy**
Slow and clean may beat fast and messy.

Track 3: Precision Challenge

Walls + obstacles. Accuracy matters.



WIN CONDITION: Fastest time + fewest collisions

GOAL + RULES

- 1 Mission**
Navigate from A to B without crashing.
- 2 Course**
Walls, turns, and obstacles.
- 3 Penalty**
Wall hits and obstacle hits count.
- 4 Debug**
Use evidence from each run to improve.

How To Win

A good run is fast, controlled, and repeatable.

TRACK 1

First Turn

- Fastest time wins
- One clean A → B run

TRACK 2

Dodge Obstacles

- Fastest time wins
- Fewest obstacle hits breaks ties

TRACK 3

Precision Challenge

- Fastest time wins
- Fewest total collisions breaks ties
- Most accurate run breaks final ties

A robot that finishes slowly is better than a robot that crashes quickly.